Community Health Clinic Queue Simulator

1. Introduction

The idea for this project originated from discussions with a family member who works in a community healthcare clinic. During these conversations, concerns were raised regarding long patient waiting times and the challenges associated with managing limited nursing staff. A common question was whether simply hiring additional nurses would solve the problem or whether staffing decisions involved more complex operational trade-offs.

Community health clinics frequently face challenges associated with limited staffing and fluctuating patient demand. Long waiting times can reduce patient satisfaction and place additional pressure on healthcare workers. Determining an appropriate number of nurses is therefore an important operational decision, as understaffing can create service bottlenecks while overstaffing can increase operational costs.

This project develops a Community Health Clinic Queue Simulator using the SimPy discrete-event simulation library. The simulator models patient arrivals, nurse availability, waiting queues, and treatment processes to investigate how staffing levels influence clinic performance. By analysing waiting times, treatment durations, and nurse utilisation, the simulation provides insight into resource allocation decisions within a healthcare environment.

2. Problem Definition

Community health clinics must balance service quality with staffing costs while operating under limited resources and variable patient demand. Insufficient nursing resources may lead to excessive patient waiting times, queue formation, and reduced service quality, while excessive staffing can result in underutilised resources and increased operating costs.

The problem addressed by this project is:

How do nurse availability and patient demand affect clinic performance, waiting times, and resource utilisation ?

The simulation aims to model these interactions and provide measurable performance indicators that can support staffing and resource allocation decisions. Through the analysis of patient waiting times, treatment durations, and nurse utilisation, the project evaluates the trade-off between resource efficiency and service quality within a community healthcare environment.

3. Societal Context

Access to timely healthcare services is an important community concern. Long waiting times can delay treatment, reduce patient satisfaction, and increase pressure on healthcare staff. Healthcare organisations must therefore make informed decisions regarding staffing levels and resource allocation.

This project demonstrates how simulation can be used to evaluate operational decisions before implementing them in real-world environments. By identifying bottlenecks and analysing resource utilisation, organisations can make evidence-based decisions that improve service quality while managing operational costs.

4. Requirements

Functional Requirements

- Accept user input for:
  - Number of nurses
  - Number of patients
  - Clinic operating duration
  - Minimum treatment time
  - Maximum treatment time
- Simulate patient arrivals.
- Simulate nurse allocation and queueing.
- Record patient statistics.
- Calculate performance metrics.
- Generate visualisations.
- Save generated graphs to an output directory.

5. Research

Discrete-event simulation is widely used in healthcare systems to analyse patient flow, waiting times, resource allocation, and service bottlenecks. SimPy provides a framework for modelling resources, queues, and existing processes, making it suitable for healthcare queue simulations.

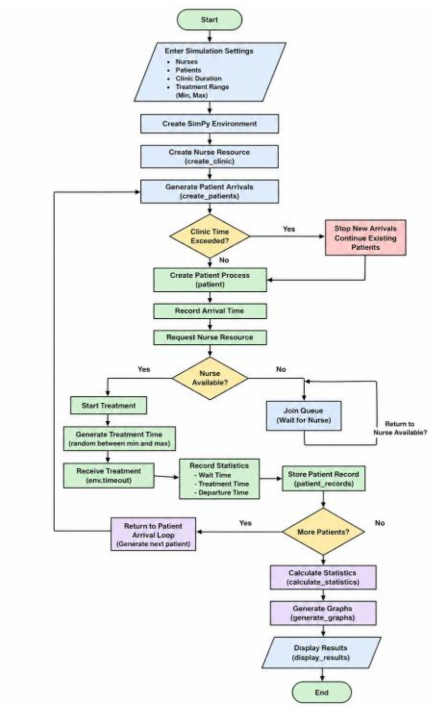
Queueing theory suggests that increasing resource utilisation often increases waiting times, particularly when utilisation approaches 100%. Therefore, highly utilised resources may become bottlenecks that reduce overall system performance.

6. Assumptions

The simulation operates under the following assumptions:

- All nurses have identical capabilities.
- Patients are served on a first-come, first-served basis.
- Treatment times are randomly generated within a specified range.
- Patients arriving before clinic closing time remain eligible for treatment.
- No patient abandons the queue.
- Emergency prioritisation is not considered.

7. System Design



8. Program Structure

The program is divided into several functions as shown below:

create_clinic()

Creates the nurse resource used within the simulation.

create_patients()

Generates patient arrivals throughout the clinic operating period.

patient()

Represents an individual patient process and records waiting and treatment statistics.

calculate_statistics()

Calculates simulation performance metrics.

display_results()

Displays simulation results to the user.

generate_graphs()

Creates visualisations showing patient wait times and patients served.

9. Results and Discussion

Example simulation configuration:

- Nurses: 1
- Patients: 10
- Clinic Duration: 120 minutes
- Treatment Time: 15–30 minutes

Results:

- Patients Served: 10
- Average Wait Time: 83.70 minutes
- Maximum Wait Time: 155.00 minutes
- Average Treatment Time: 23.10 minutes
- Nurse Utilisation: 97.88%

The results indicate that a single nurse becomes a bottleneck within the system. Although utilisation is extremely high, patient waiting times increase significantly due to queue formation. This demonstrates that maximising utilisation does not necessarily maximise service quality.

10. Conclusion

This project successfully demonstrates how discrete-event simulation can be used to analyse healthcare operations. The simulation highlights the relationship between staffing levels, waiting times, and resource utilisation. Results show that excessive resource utilisation can create bottlenecks and negatively impact patient service levels. Future improvements may include patient prioritisation, multiple treatment categories, appointment scheduling, and additional healthcare resources.

References

https://www.tandfonline.com/doi/epdf/10.1057/palgrave.jors.2600669?needAccess=true

https://simpy.readthedocs.io/en/latest/topical_guides/resources.html

https://www.linkedin.com/pulse/queuing-theory-science-managing-flow-through-dd5qc/